

SFPPy Syntax: Operators and in Migration Simulations

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Introduction

1. The Operator: Layer Combination and Result Merging

1.1. Assembling Multilayer Packaging 

1.2. Merging Simulation Results 

2. The Operator: Chaining Migration Steps

2.1. General Usage 

2.2. Handling Multiple Food Contact Conditions 

2.3. Updating Properties Mid-Simulation 

3. Example: Advanced Simulation Workflow

Conclusion 

Introduction

SFPPy enables sophisticated mass transfer simulations in food packaging systems using a compact, Pythonic syntax. This page details the use of two key operators:

-  (Addition Operator) : For combining layers or results
-  (Right Shift Operator) : For chaining migration simulations

These operators simplify simulations while maintaining full control over the process.

Note: the transfer of food contact conditions `food >> material` can be also represented as `food @ material`.

1. The Operator: Layer Combination and Result Merging

1.1. Assembling Multilayer Packaging

#

In SFPPy, packaging materials are modeled as **layers**, which can be combined using the  operator.

```
from patankar.layer import gPET, PP

A = gPET(l=(20, "um")) # 20 µm PET
B = PP(l=(500, "um")) # 500 µm PP

ABA = A + B + A # Creates a trilayer structure (ABA)
```

1.2. Merging Simulation Results

#

After running simulations for different food contact conditions, results can be **combined** using `+`.

```
fullsolution = foodcontact1.lastsimulation + foodcontact2.lastsimulation
```

This aggregates migration profiles from multiple steps into a single dataset.

2. The Operator: Chaining Migration Steps

2.1. General Usage

#

The `>>` operator simplifies **sequential simulations**, ensuring automatic propagation of **geometry, temperature, and mass transfer states**.

```
mypackaging >> myfood >> mymaterial >> myfood
```

or equivalently

```
mypackaging @ myfood >> mymaterial >> myfood
```

This executes:

1. **Geometry matching** between `mypackaging` and `myfood`
2. **Temperature propagation** to `mymaterial`
3. **Mass transfer simulation** into `myfood`

2.2. Handling Multiple Food Contact Conditions

#

For sequential food contact scenarios:

```
mypackaging >> foodcontact2 >> foodcontact1 >> mymaterial >> foodcontact1 >> foodcontact2
```

This ensures:

- **Geometry adaptation** for both food contacts
- **Thermal equilibration**
- **Mass transfer simulation**
- **State propagation** across conditions

Each step stores results, accessible via:

```
foodcontact1.lastsimulation  
foodcontact2.lastsimulation
```

To explicitly store the final result:

```
result = ... step n >> step n+1
```

2.3. Updating Properties Mid-Simulation

#

SFPPy allows **dynamic property updates** while chaining simulations:

```
foodcontact1 >> mymaterial.update(substance="new migrant", l=new_thickness) >> foodcontact1
```

3. Example: Advanced Simulation Workflow

Consider a **trilayer (ABA)** packaging with **limonene** migration:

```
from patankar.food import fat, ambient, hotfilled
from patankar.loadpubchem import loadpubchem

# Define the migrant
m = loadpubchem('limonene')

# Define packaging structure
A = gPET(l=(20, "um"))
B = PP(l=(500, "um"), migrant=m, CP0=200)
ABA = A + B + A

# Define food contact conditions
medium1 = ambient(contacttemperature=(20, "degC"), contacttime=(4, "months"))
medium2 = hotfilled(contacttemperature=(90, "degC"))
medium3 = fat(contacttemperature=(30, "degC"), contacttime=(6, "months"))

# Simulate migration
medium1 >> ABA >> medium1 >> medium2 >> medium3

# Combine results
sol123 = medium1.lastsimulation + medium2.lastsimulation + medium3.lastsimulation
sol123.plotCF()
```

This example demonstrates how **minimal syntax** manages complex simulation steps efficiently. ⚡

Conclusion

SFPPy's **+** and **>>** operators provide a powerful, intuitive syntax for modeling **mass transfer simulations** in food packaging systems. By leveraging these operators, users can **reduce code complexity while preserving flexibility**. 🏆



SFPPy for Food Contact Compliance and Risk Assessment

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